

Practical Data Science (with Python) Assignment 2

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Introduction

- This report works to understand wine quality assessment, a dataset that includes important factors that make up different wines. Our aim is to build a predictive model that can estimate wine quality based on these attributes. In order to achieve this, we explored machine learning, fine-tuned parameters of clustering, and analysed their performance. Our dataset includes features such as fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulphur dioxide, total sulphur dioxide, density, pH, sulphates, and alcohol. These attributes will be measured to see if they influence the overall quality of wine. The modelling efforts will aim to uncover the relationships between these factors and wine quality.

Data Preparation and Analysis

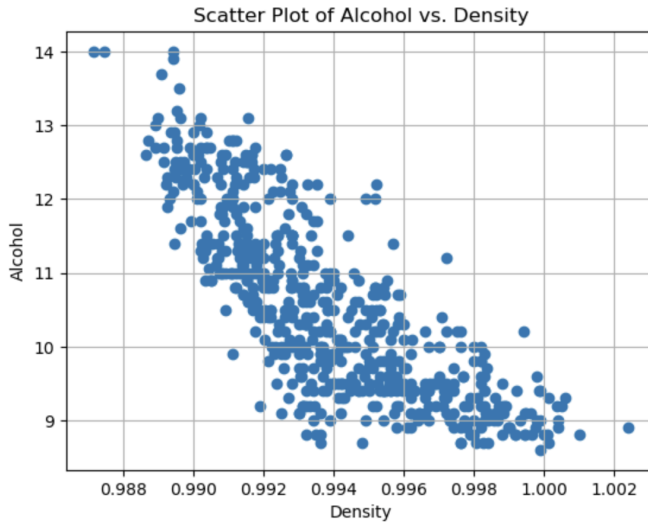
Data retrieval

- We have retrieved the data ensuring that we have the right headers in place separating the values which were originally separated using “;”. Next 600 from the dataset were randomly selected to act as our “A2RandomSample.csv”. We ensured that we got no missing values in our dataset by first dropping all missing values, this method is simple and since we have many to choose from it does not effect our analysis.

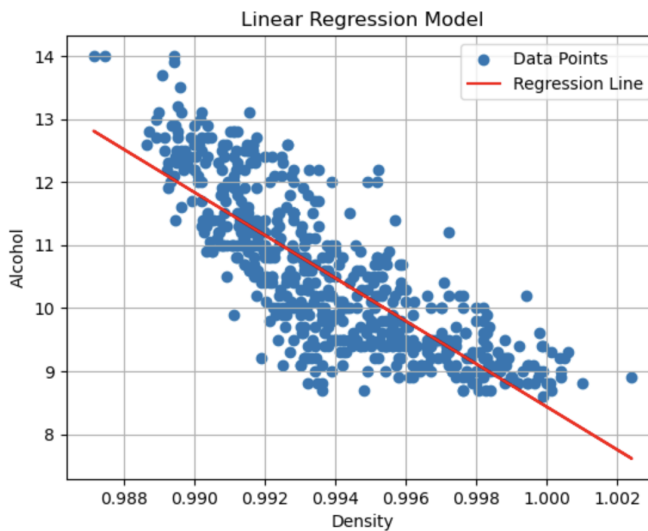
	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
0	6.8	0.34	0.44	6.6	0.052	28.0	156.0	0.99550	3.14	0.41	9.6	5.0
1	8.3	0.21	0.41	2.2	0.050	24.0	108.0	0.99400	2.85	0.45	9.5	5.0
2	7.2	0.19	0.31	1.4	0.046	37.0	135.0	0.99390	3.34	0.57	10.2	7.0
3	6.6	0.38	0.29	2.4	0.136	15.0	93.0	0.99336	3.18	0.60	9.5	5.0
4	6.5	0.24	0.39	17.3	0.052	22.0	126.0	0.99888	3.11	0.47	9.2	6.0

the relationship between two variables:alcohol and density

- Below is a scatter plot showing the relationship between alcohol and density. Looking at the data points on the scatter plot it can be observed that it has a strong linearity, in particular it has a strong negative correlation. This means that higher the alcohol concentration the lower the density. Through the4 below plot, density does not have a high range where our graph records values from 0.986 to 1.004.



- Below is a linear regression model where we have the line of best fit running through the scatter plot. The line reinforces what was stated above showing the clear negative correlation between alcohol and density. This line was later calculated through $y=mx+b$ where our least squares equation came out to $Y = -340.0843X + 348.5167$.



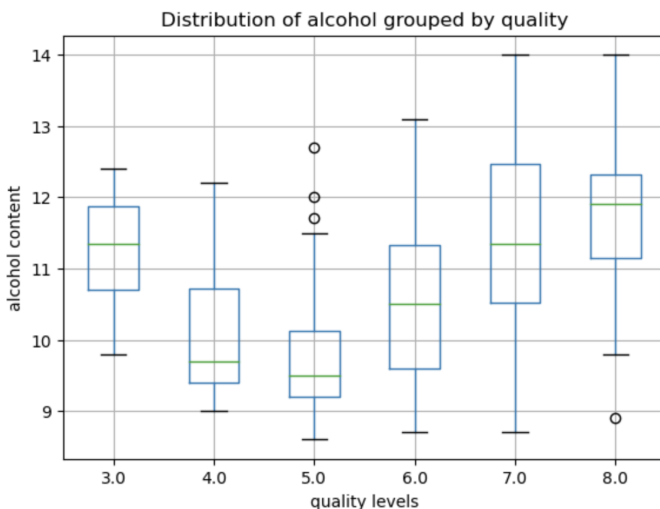
the relationship between two variables: quality and alcohol

- Using the code value counts it was identified that within the sample dataset the column quality has 6 unique groups. Given this we can expect to have 6 different box plots when comparing alcohol to the different quality groups. The different quality groups going from 3 to 8 show scores between 0 (very bad) and 10 (very excellent) thus for our dataset 3 is the lowest quality group and 8 is the highest.

Unique groups	frequency
6	280
5	168

7	98
4	26
8	24
3	4

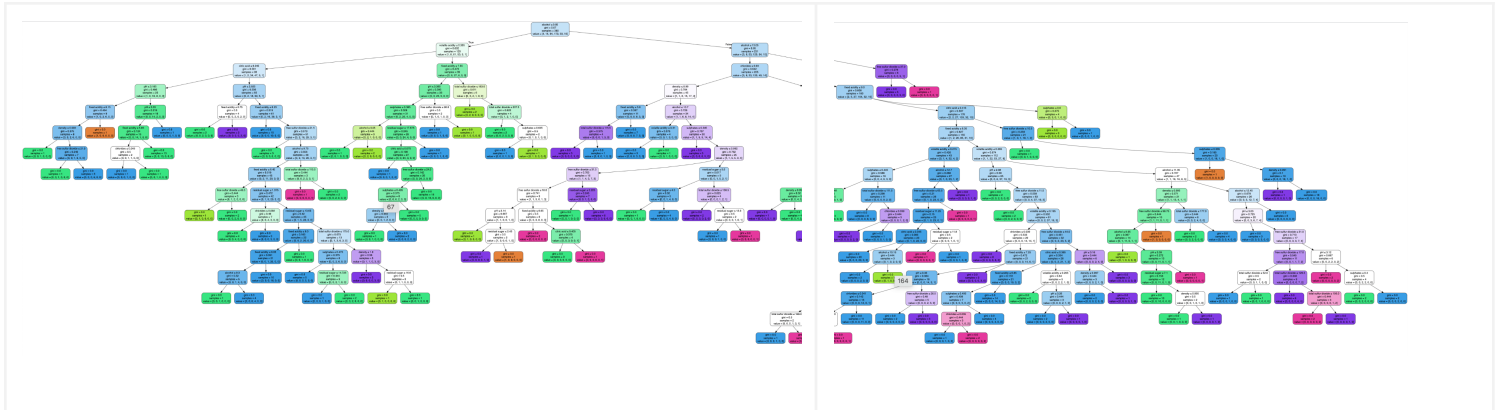
- Below are a series of boxplots of alcohol content grouped by the different quality levels. The purpose is to check whether there is correlation between the different quality levels and alcohol content. Looking at the median values for each box plot the two groups appear to have no correlation where the median values are quite scattered with quality level 5 which resides in the middle having the lowest followed closely by quality group 4. Another observation is that the ranges of the groups greatly differ where quality group 7 has an overwhelming range compared to quality group 3 at a range of about “5.5” compared to a range of about “3”. A couple of outlets are present within quality groups 5 and 8. Overall the box plots do not show any clear sign of correlation, where wine of low quality levels as well as wine with high quality levels are shown to have high alcohol content.



Classification

Decision tree

- Below is a decision tree. A decision tree is a flow chart-like structure in which each internal node represents a test on an attribute. Each branch represents the outcome of a test, and each leaf node represents a class label. The path from the root to leaf represents classification rules.



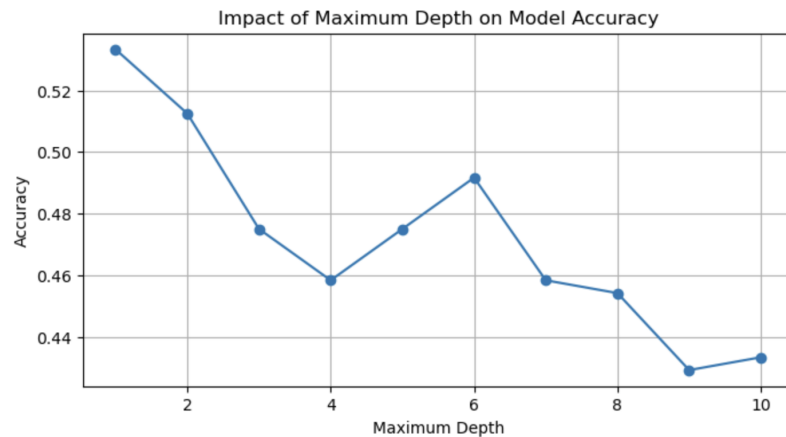
- Below is the classification report outputted by our code in python on the left are the different wine qualities followed by a series of metrics which calculate how effective the classification or predictive power is. We have precision, recall, f1 score and support.

	precision	recall	f1-score	support
3.0	0.00	0.00	0.00	0
4.0	0.27	0.25	0.26	12
5.0	0.55	0.53	0.54	74
6.0	0.45	0.42	0.44	107
7.0	0.26	0.28	0.27	39
8.0	0.10	0.12	0.11	8
accuracy			0.41	240
macro avg	0.27	0.27	0.27	240
weighted avg	0.43	0.41	0.42	240

- Precision is the measure of accuracy of positive predictions, indicating how often the model correctly predicts each quality class. The precision values vary across different quality classes, with the quality of 5.0 achieving the highest precision at 0.55. This suggests that the model's predictions for it are accurate about 55% of the time. Recall is the measure of the model's ability to identify all relevant instances of each quality class. The recall values indicate the model's effectiveness in capturing instances of each quality class. For instance, the model correctly identifies 53% of instances of class 5.0, as shown by the recall score of 0.53. The F1-Score is the average of precision and recall, providing a measure of model performance. It takes into account both false positives and false negatives. The F1-Score values reflect the model's overall performance in classifying wine quality. Class 5.0 achieves the highest F1-Score at 0.54, indicating a balanced performance in terms of precision and recall. Overall the model is not very strong at all where at most the model is only 55% for precision, 53% for recall and 54% for f1- score. With many scores being far lower than these it is hard to tell whether the model is able to make accurate predictions.

2.2

- One key parameter of the model is the max depth, in the original decision tree we did not include a max depth however we had now experimented with different levels. Below is a graph measuring accuracy with the different max depths. It would seem that none, 1 and 2 have high accuracy which they do, however it is not just the accuracy we are looking for. Also with this graph it was observed that a max depth of 6 has a higher accuracy proving that a higher max depth does not lower the models accuracy.



- Below are a series of reports evaluating the models with different max depths, in the following order 2, 4, 6, 8. With the charts shown the most suitable max depth value to obtain the best model is one with a max depth score of 2. This is as having a max Depth of 2 allows for a slightly deeper tree with two levels of decision nodes. The positives are that it is relatively balanced between simplicity and complexity, potentially offering a good trade-off. For a quality of 5.0, it has a relatively high precision 0.70 but a lower recall 0.31 and F1-score 0.43. This indicates that the model is decent at identifying instances of class 5.0 but might miss some. For quality 6.0, it has a moderate precision 0.49 and a high recall 0.92, resulting in a relatively high F1-score 0.63. The model performs well in identifying instances of class 6.0. For quality 4.0 and 8.0, precision, recall, and F1-scores are all low, indicating that the model struggles to correctly classify instances of these classes. For quality 7.0, the model has low precision 0.40, low recall 0.05, and a very low F1-score 0.09, suggesting significant difficulty in identifying instances for quality 7.0.

	precision	recall	f1-score	support
4.0	0.00	0.00	0.00	12
5.0	0.70	0.31	0.43	74
6.0	0.49	0.92	0.63	107
7.0	0.40	0.05	0.09	39
8.0	0.00	0.00	0.00	8
accuracy			0.51	240
macro avg	0.32	0.26	0.23	240
weighted avg	0.50	0.51	0.43	240

	precision	recall	f1-score	support
4.0	0.17	0.08	0.11	12
5.0	0.71	0.34	0.46	74
6.0	0.46	0.72	0.56	107
7.0	0.22	0.18	0.20	39
8.0	0.00	0.00	0.00	8
accuracy			0.46	240
macro avg	0.31	0.26	0.27	240
weighted avg	0.47	0.46	0.43	240

	precision	recall	f1-score	support
4.0	0.60	0.25	0.35	12
5.0	0.75	0.41	0.53	74
6.0	0.48	0.73	0.58	107
7.0	0.23	0.18	0.20	39
8.0	0.00	0.00	0.00	8
accuracy			0.49	240
macro avg	0.41	0.31	0.33	240
weighted avg	0.51	0.49	0.47	240

	precision	recall	f1-score	support
3.0	0.00	0.00	0.00	0
4.0	0.25	0.17	0.20	12
5.0	0.49	0.50	0.49	74
6.0	0.47	0.60	0.53	107
7.0	0.33	0.15	0.21	39
8.0	0.00	0.00	0.00	8
accuracy			0.45	240
macro avg	0.26	0.24	0.24	240
weighted avg	0.43	0.45	0.43	240

2.3

- Below are the different metrics for different train/ test split ratios going from 2:8 to 8:2. While comparing the different results from retrasing the dataset with the different splits, we can tell that 0.3 or 3:7 ratio and 0.4 or 4:6 ratio are the best. The numbers are lower for training sets that are below the 3:7 ratio however for a ratio greater than 4:6 we were not able to predict well.

Split Ratio: 0.2				
	precision	recall	f1-score	support
3.0	0.00	0.00	0.00	4
4.0	0.23	0.14	0.17	22
5.0	0.30	0.35	0.32	131
6.0	0.49	0.61	0.55	226
7.0	0.33	0.13	0.19	77
8.0	0.00	0.00	0.00	20
accuracy			0.41	480
macro avg	0.23	0.20	0.20	480
weighted avg	0.38	0.41	0.38	480

Split Ratio: 0.3				
	precision	recall	f1-score	support
3.0	0.00	0.00	0.00	4
4.0	0.30	0.18	0.22	17
5.0	0.26	0.29	0.28	110
6.0	0.49	0.63	0.55	197
7.0	0.27	0.11	0.16	72
8.0	0.00	0.00	0.00	20
accuracy			0.40	420
macro avg	0.22	0.20	0.20	420
weighted avg	0.35	0.40	0.37	420

Split Ratio: 0.4				
	precision	recall	f1-score	support
3.0	0.00	0.00	0.00	4
4.0	0.29	0.14	0.19	14
5.0	0.25	0.30	0.27	94
6.0	0.48	0.61	0.54	173
7.0	0.33	0.12	0.17	59
8.0	0.00	0.00	0.00	16
accuracy			0.40	360
macro avg	0.23	0.20	0.20	360
weighted avg	0.36	0.40	0.37	360

Split Ratio: 0.5				
	precision	recall	f1-score	support
3.0	0.00	0.00	0.00	4
4.0	0.20	0.08	0.12	12
5.0	0.30	0.39	0.34	76
6.0	0.50	0.61	0.55	145
7.0	0.42	0.16	0.23	51
8.0	0.00	0.00	0.00	12
accuracy			0.42	300
macro avg	0.24	0.21	0.21	300
weighted avg	0.40	0.42	0.40	300

Split Ratio: 0.6				
	precision	recall	f1-score	support
3.0	0.00	0.00	0.00	4
4.0	0.00	0.00	0.00	7
5.0	0.26	0.35	0.30	60
6.0	0.50	0.61	0.55	119
7.0	0.43	0.08	0.13	39
8.0	0.00	0.00	0.00	11
accuracy			0.40	240
macro avg	0.20	0.17	0.16	240
weighted avg	0.38	0.40	0.37	240

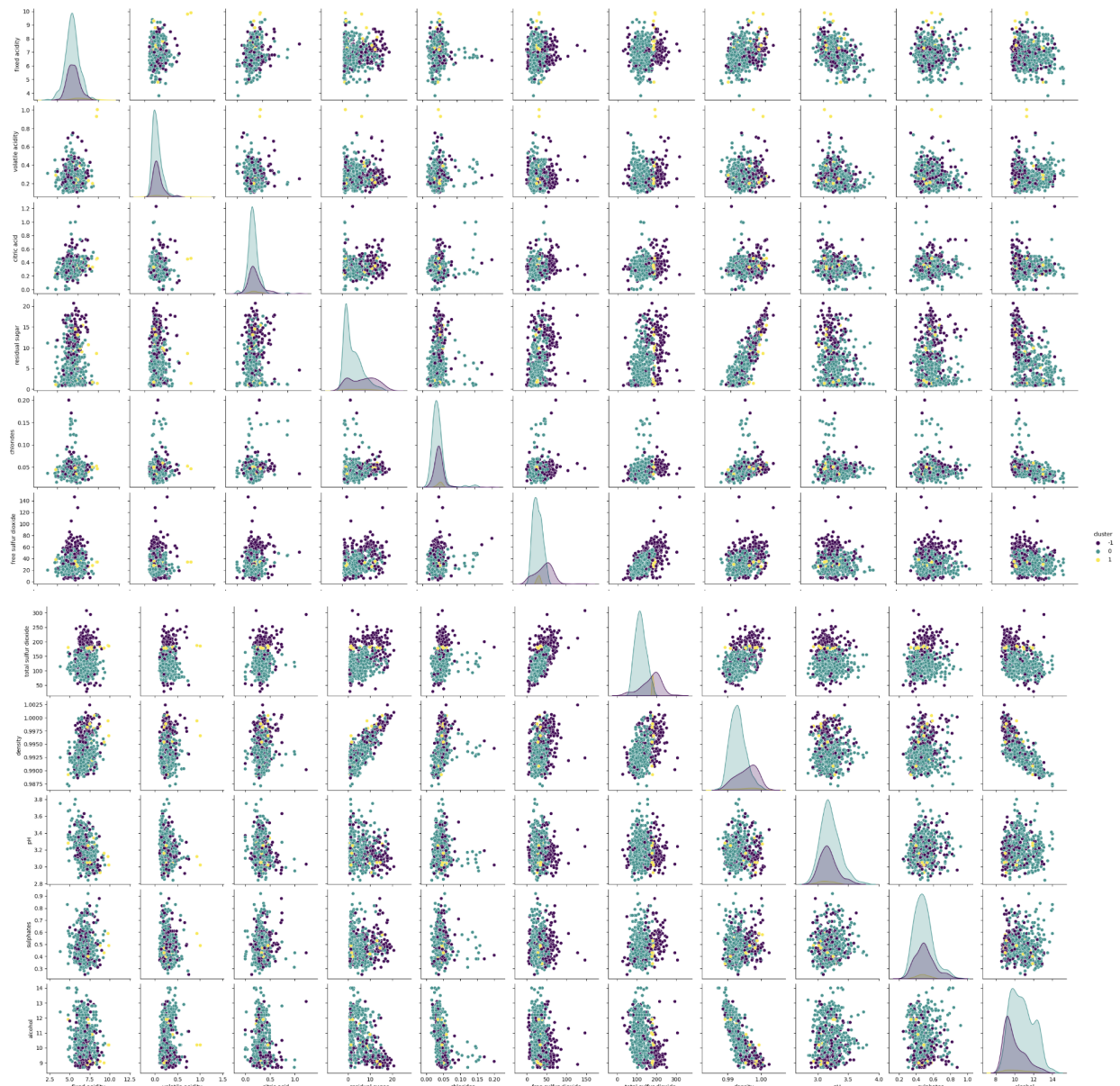
Split Ratio: 0.7				
	precision	recall	f1-score	support
3.0	0.00	0.00	0.00	3
4.0	0.00	0.00	0.00	4
5.0	0.31	0.39	0.34	46
6.0	0.50	0.61	0.55	89
7.0	0.18	0.06	0.10	31
8.0	0.00	0.00	0.00	7
accuracy			0.41	180
macro avg	0.16	0.18	0.16	180
weighted avg	0.36	0.41	0.38	180

Split Ratio: 0.8				
	precision	recall	f1-score	support
3.0	0.00	0.00	0.00	2
4.0	0.00	0.00	0.00	2
5.0	0.24	0.39	0.30	31
6.0	0.47	0.51	0.49	59
7.0	0.17	0.05	0.07	22
8.0	0.00	0.00	0.00	4
accuracy			0.36	120
macro avg	0.15	0.16	0.14	120
weighted avg	0.32	0.36	0.33	120

Clustering

3.1

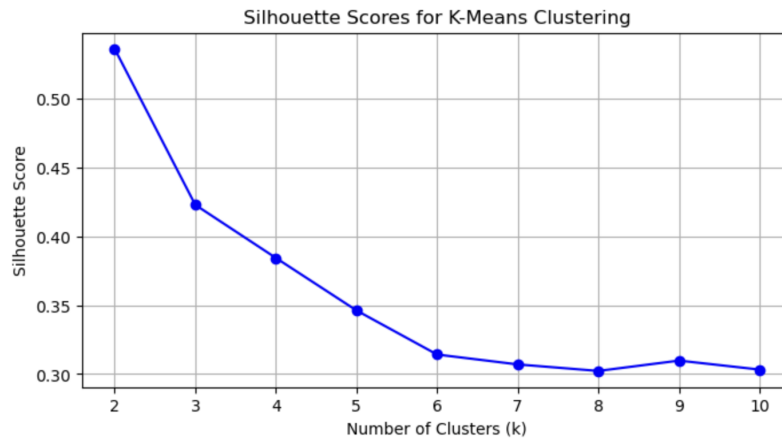
- The method of clustering chosen was DBSCAN, this method was chosen because of its ability to identify clusters of arbitrary shapes . Below is the clustered data. We have three clusters obtained using an eps value of 8 and a min sample value of 10. These values were chosen to best display the different clusters within the dataset. In more detail eps of 8 was chosen after testing using an elbow graph . The min sample of 10 was chosen because we wanted clusters to have significant meaning especially for a larger dataset we don't want too many individual clusters appearing.



3.2

The optimal number of clusters is believed to be 2, this is where our data is relatively focused on a central point, being quite dense. It would be meaningless to have only 1 cluster where it does not show much analysis. However having too many takes away from the credibility of each cluster as they would consist of few points. we wanted substantial clusters thus it was believed that 2 was most suitable. Below is a graph which we used to help identify the right amount of clusters. To read this we look at the point where the graph starts to level out, as can be seen point 3 which lies in between point 2 and 4 is whether the line starts to level out. This graph suggests that adding more clusters beyond point 3 doesn't significantly improve the clustering quality. The concept behind the graph is that it helps make an informed decision about the number of clusters to use in K-Means clustering analysis. The optimal k is the number of clusters that

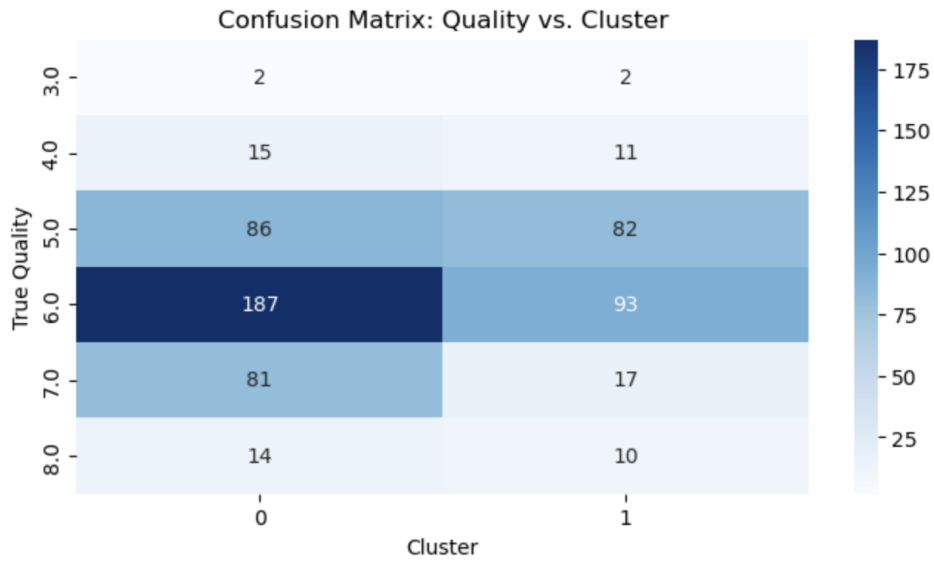
maximises the Silhouette Score, indicating better clustering quality. Reading this graph we see that the optimal number of clusters is 2 and the optimal silhouette score is 0.54



Optimal number of clusters (k): 2
Optimal Silhouette Score: 0.54

3.3

- Below is the outputted confusion matrix, it shows the clusters that we have and the number of points in each regarding quality levels. The columns represent the different clusters and the rows the true quality levels. We once again see that we have 2 clusters. The y axis on the left shows the quality levels, the scale on the right shows the number of points that were located in a certain quality where deeper the blue the more points. The x axis is the 2 clusters being whether 0 or 1. Looking at this confusion matrix most of their values are centred around quality 6 and cluster 0 with an overwhelming number of data points. The overall analysis of this when comparing the clusters is that Cluster 0 appears to be more diverse, covering a wider range of True Quality levels. Whereas cluster 1 seems to have a narrower focus, with fewer data points for most True Quality levels. The distribution of True Quality within the clusters implies that Cluster represents a broader range of wine quality, while Cluster 1 may represent a more specific subset of quality levels.



Reference

All coding was done on Jupyter note - Kluyver, T. et al., 2016. Jupyter Notebooks

Ren, y. (2022). Lecture Notes. Practical data science (COSC2670). October 1, 2023.

Ren, y. (2022). La Notes. Practical data science (COSC2670). October 1, 2023.